

Project Design Phase-II

Data Flow Diagram & User Stories

Date	28 September 2026
Team ID	SWTID-2026-2313
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Data Flow Diagram (Level 0)

External Entities: API Client (User/Reader) and the Google Gemini AI service.

Process: The Express.js server (Auth, Blog, and AI controllers).

Data Store: MongoDB — Users collection and Blogs collection.

- API Client → [HTTP Request: register/login/CRUD/AI] → Express Server
- Express Server → [Read/Write User & Blog documents] → MongoDB
- Express Server → [Prompt: topic or content] → Gemini AI → [Generated text] → Express Server
- Express Server → [HTTP Response: JSON] → API Client

AI BLOGNEST API — DATA FLOW DIAGRAM (LEVEL 0)

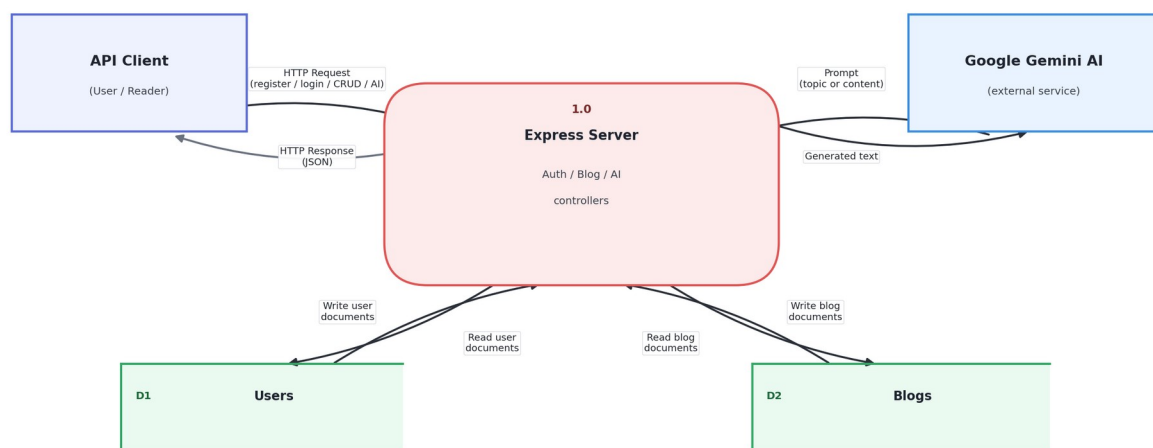


Figure 1: High-level data flow through AI BlogNest API

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Author (Registered User)	Registration	USN-1	As a user, I can register for the application	I receive a 201 response with my	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
			by entering my name, email, and password.	new user id, name, and email.		
Author (Registered User)	Login	USN-2	As a user, I can log into the application using my email & password.	I receive a signed JWT token I can use for authenticated requests.	High	Sprint-1
Author (Registered User)	Profile	USN-3	As a user, I can view my own profile.	I receive my id, name, and email when authenticated.	Medium	Sprint-1
Author (Registered User)	Blog Management	USN-4	As an author, I can create a new blog post.	The blog is saved with my user id as the author and returned to me.	High	Sprint-2
Author (Registered User)	Blog Management	USN-5	As an author, I can update or delete only my own blog posts.	I receive a 403 error if I try to edit a post I did not author.	Medium	Sprint-2
Reader (Public/Guest)	Blog Management	USN-6	As a reader, I can view all blog posts without an account.	I receive a list of blogs with author name populated.	High	Sprint-2
Reader (Public/Guest)	Blog Management	USN-7	As a reader, I can view a single blog post by its ID.	I receive the full blog post or a 404 if it doesn't exist.	Medium	Sprint-2
Author	AI Content	USN-8	As an	A	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
(Registered User)	Generation		author, I can generate a blog draft from a topic.	structured blog post is generated, categorized, and saved under my account.		
Author (Registered User)	AI Summarization	USN-9	As an author, I can summarize any block of content.	I receive a short, readable paragraph summarizing the input.	High	Sprint-4